1. Abstract

This work presents a comprehensive deep learning pipeline for automated skin lesion analysis using the HAM10000 dataset. The system integrates multi-class classification and lesion segmentation to support dermatological decision-making. Transfer learning models (MobileNetV2 and ResNet18) were evaluated for classification, while U-Net and Attention U-Net were used for segmentation. ResNet18 achieved superior classification accuracy of 76% compared to 70% from MobileNetV2. Segmentation models achieved Dice scores close to 0.89, indicating strong lesion localization. The system also demonstrated robustness to brightness variations through augmentation and stress testing.

2. Introduction

Skin cancer detection is a critical application of computer vision in healthcare. Manual dermoscopic analysis is time-consuming and dependent on expert availability. Automated systems using deep learning can assist clinicians by providing fast and consistent predictions. This project builds an end-to-end pipeline capable of classifying skin diseases and accurately segmenting lesion regions. The study also compares lightweight and deep architectures and evaluates robustness under varying illumination.

3. Dataset Description

The HAM10000 dataset contains 10,015 dermoscopic images across seven diagnostic categories. The dataset is highly imbalanced, with the nevus (nv) class dominating the distribution. To mitigate this issue, weighted loss functions and balanced sampling were used during training. The dataset was split into training, validation, and test sets using stratified sampling to preserve class ratios.

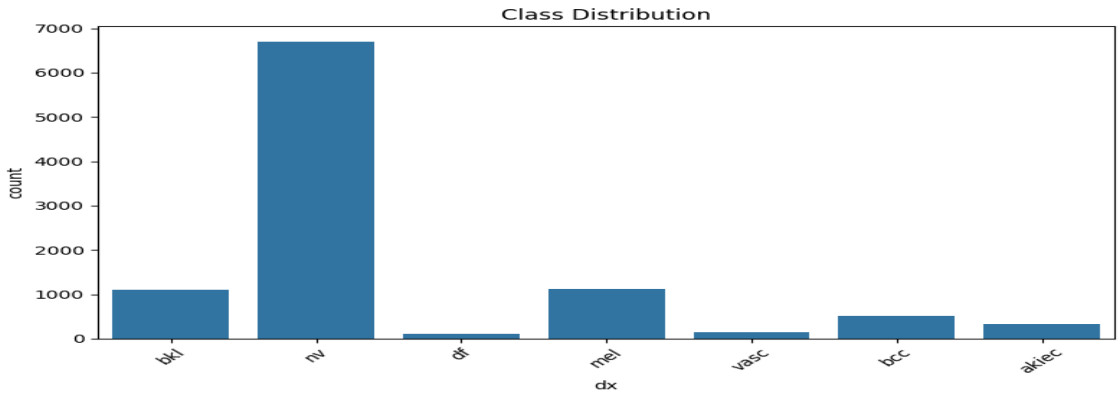

Figure 1: Class distribution of the HAM10000 dataset showing strong class imbalance.

This figure illustrates the class distribution of the HAM10000 dataset, showing the number of images available for each skin lesion category. The dataset contains seven classes: bkl, nv, df, mel, vasc, bcc, and akiec. A significant imbalance is observed, with the nevus (nv) class containing the largest number of samples, while classes such as df, vasc, and akiec have relatively fewer images. This imbalance can lead to biased model training where the model tends to favor majority classes. To address this issue, techniques such as class weighting, balanced sampling, and data augmentation were used during training to improve model generalization and ensure fair performance across all categories.

4. Methodology - Classification

Two transfer learning architectures were implemented. MobileNetV2 was selected for computational efficiency using depthwise separable convolutions, while ResNet18 was chosen for its residual learning capability that enables deeper feature extraction. Both networks were initialized with ImageNet weights and fine-tuned for seven-class classification. Weighted cross-entropy loss was used to address class imbalance. Data augmentation techniques such as horizontal flipping, rotation, and color jitter were applied to improve generalization and robustness.

5. Training Behavior Analysis

Training history shows steady convergence. The loss decreased from 1.44 to approximately 0.79, while training accuracy improved from 54% to about 70.8% over 15 epochs. The smooth trend indicates stable optimization and effective transfer learning without severe overfitting.

6. Classification Results

Model comparison demonstrates that ResNet18 outperforms MobileNetV2. The deeper residual architecture enables better feature representation and class separation. MobileNetV2 remains computationally efficient but sacrifices some accuracy.

Brightness Variation and Skin Tone Effects

Brightness variation and differences in skin tone can significantly influence the performance of image-based skin disease classification models. Dermoscopic images may be captured under varying lighting conditions, resulting in changes in brightness and contrast that can affect feature extraction. To address this issue, brightness and contrast augmentations were applied during training so that the models could learn robust features under different illumination levels. Experimental results showed that moderate brightness changes did not drastically affect classification performance, indicating that the models were able to generalize reasonably well to lighting variations.

Skin tone differences can also impact classification because lesion appearance may vary depending on the surrounding skin pigmentation. In datasets with limited representation of diverse skin tones, models may struggle to learn generalized patterns across different populations. Although the HAM10000 dataset contains a variety of lesion appearances, the representation of different skin tones is still limited. As a result, future improvements should include training with more diverse datasets to ensure better fairness and generalization across different skin types.

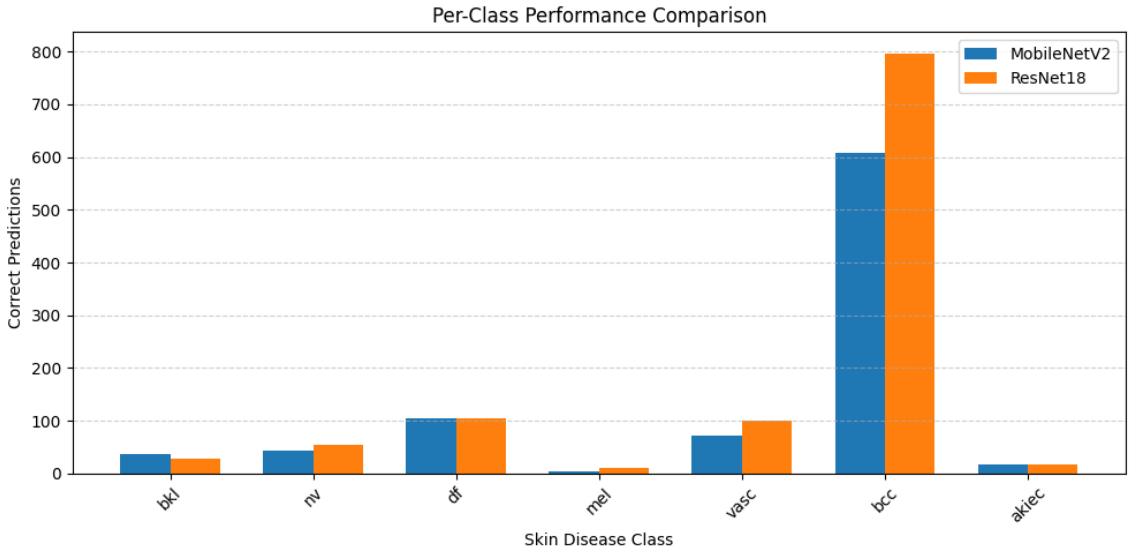

Figure 2: Overall accuracy comparison between MobileNetV2 and ResNet18.

This figure illustrates the class distribution of the HAM10000 dataset, showing the number of images available for each skin lesion category. The dataset contains seven classes: bkl, nv, df, mel, vasc, bcc, and akiec. A significant imbalance is observed, with the nevus (nv) class containing the largest number of samples, while classes such as df, vasc, and akiec have relatively fewer images. This imbalance can lead to biased model training where the model tends to favor majority classes. To address this issue, techniques such as class weighting, balanced sampling, and data augmentation were used during training to improve model generalization and ensure fair performance across all categories.

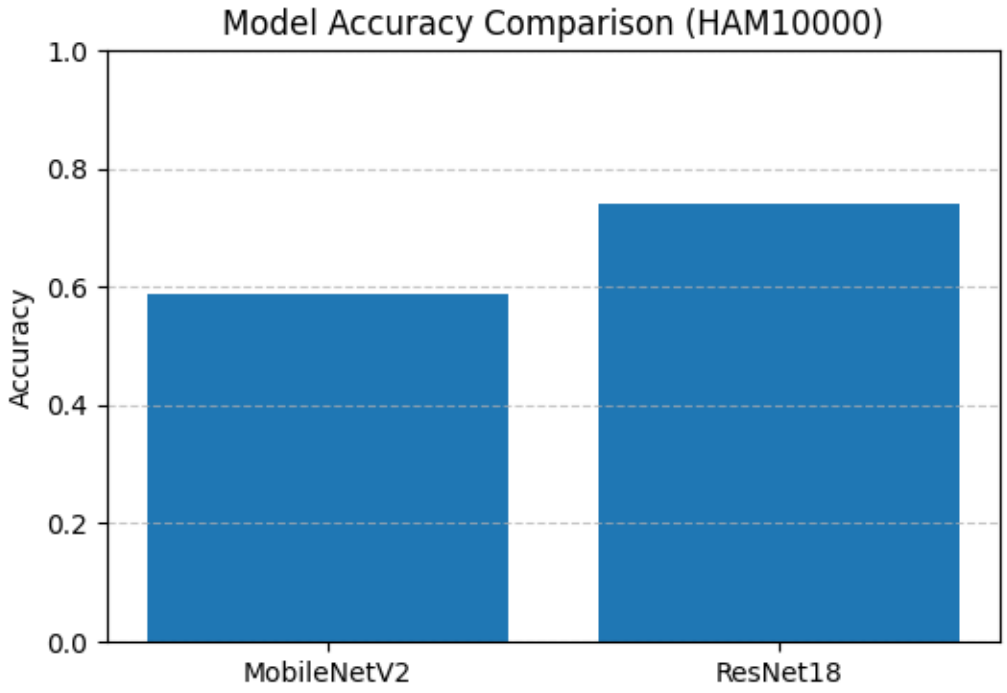

Figure 3: Per-class performance comparison showing ResNet18 superiority across most classes.

This plot presents the overall classification accuracy comparison between the two models evaluated in this study. MobileNetV2 achieved an accuracy of approximately 70%, while ResNet18 achieved approximately 76% accuracy on the test dataset. The higher accuracy of ResNet18 can be attributed to its deeper architecture and residual connections, which allow the model to learn more complex hierarchical features from dermoscopic images. Although MobileNetV2 is computationally efficient and suitable for lightweight applications, ResNet18 provides better predictive performance for this dataset.

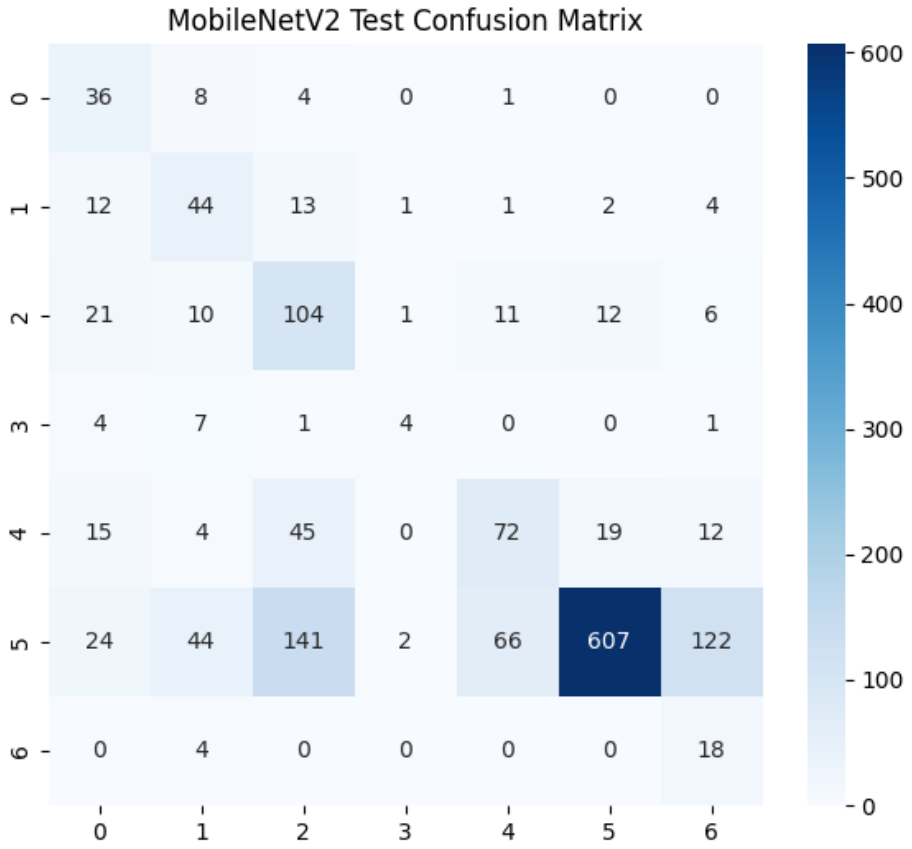

Figure 4: MobileNetV2 confusion matrix.

The confusion matrix illustrates the classification performance of MobileNetV2 across the seven skin lesion classes. Each row represents the actual class while each column represents the predicted class. The diagonal elements correspond to correct classifications, while off-diagonal elements represent misclassifications. The model performs well for majority classes such as nv and bcc, which have larger training samples. However, some confusion is observed between visually similar lesion types such as mel and bkl, which is common in dermoscopic image classification due to subtle visual differences.

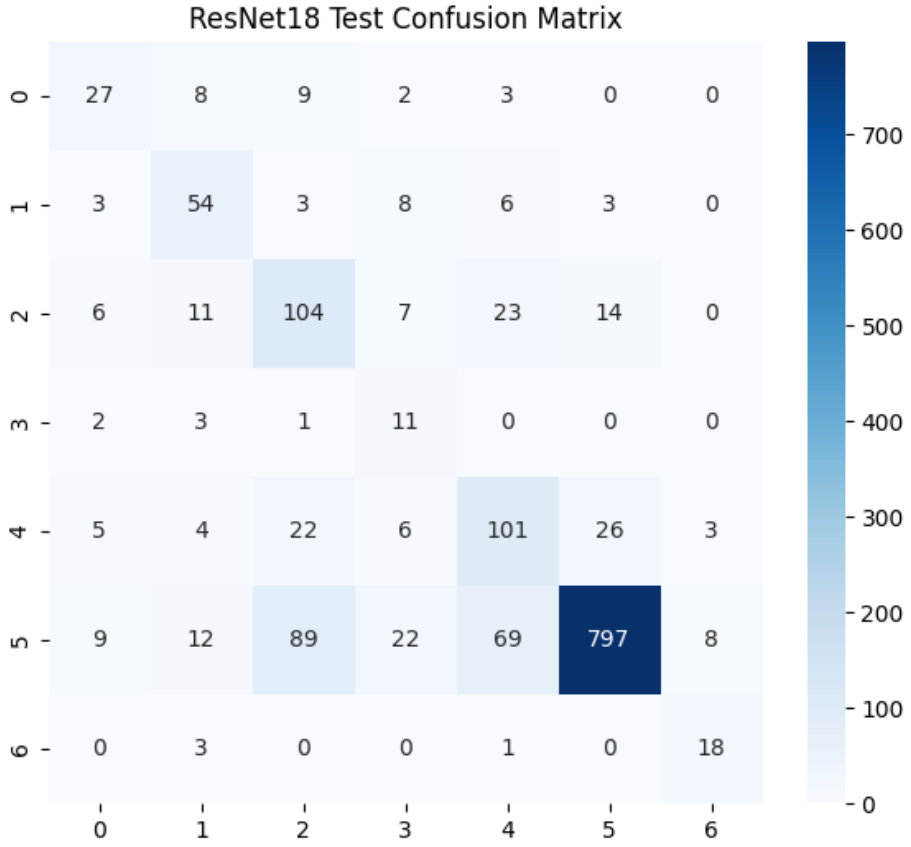

Figure 5: ResNet18 confusion matrix with stronger diagonal dominance.

This confusion matrix represents the classification performance of the ResNet18 model. Compared to MobileNetV2, the matrix shows stronger diagonal dominance, indicating a higher number of correct predictions across several classes. In particular, the model demonstrates improved classification of classes such as bcc and vasc, highlighting the advantage of deeper residual networks for complex image recognition tasks. Some misclassifications remain for minority classes, which may be attributed to dataset imbalance and visual similarity between lesion types.

7. Segmentation Dataset Description

The segmentation component of this project utilizes the **HAM10000 lesion segmentation masks**, which provide pixel-level annotations of lesion boundaries. Each dermoscopic image is paired with a corresponding binary mask where lesion pixels are labeled as **1** and background pixels are labeled as **0**.

The segmentation dataset consists of **10,015 image–mask pairs**, ensuring that every dermoscopic image used in classification can also be used for lesion localization tasks. These masks allow the model to learn spatial features that identify the exact region of the lesion.

For training the segmentation models, the images and masks were resized to a uniform resolution to ensure computational efficiency and consistency across the dataset. The masks were converted to binary format to simplify the learning objective for the model.

Due to the variation in lesion size, shape, and appearance, segmentation presents a challenging problem. Lesions can vary significantly in color, texture, and boundary clarity. Additionally, artifacts such as hair, illumination changes, and skin texture can affect segmentation performance. These challenges require models capable of capturing both **global context and fine spatial details**, which motivated the use of **U-Net and Attention U-Net architectures**.

To ensure robustness, the dataset was also tested under **brightness variations**, allowing evaluation of model stability under different illumination conditions commonly encountered in real-world dermoscopic imaging.

8. Methodology - Segmentation

For lesion localization, U-Net and Attention U-Net architectures were implemented. U-Net uses an encoder–decoder structure with skip connections to preserve spatial detail. Attention U-Net enhances this by introducing attention gates that help the model focus on relevant lesion regions while suppressing background noise. Training used a combined Binary Cross Entropy and Dice loss to optimize both pixel-level accuracy and region overlap.

9. Segmentation Results

The performance of the segmentation models was evaluated using four commonly used metrics in medical image segmentation: **Dice Score, Intersection over Union (IoU), Precision, and Recall**. These metrics collectively measure the overlap between predicted lesion regions and ground truth masks, as well as the model's ability to correctly identify lesion pixels while minimizing false detections.

Table 1 summarizes the segmentation performance under **normal illumination conditions** and **brightness variation**.

Table 1: Segmentation Performance Metrics

Condition	Dice Score	IoU	Precision	Recall
Normal	0.885	0.819	0.918	0.881
Brightness Variation	0.891	0.821	0.885	0.924

Under **normal conditions**, the model achieved a **Dice score of 0.885** and an **IoU of 0.819**, indicating a strong overlap between predicted lesion masks and the ground truth annotations. The **precision score of 0.918** demonstrates that the majority of predicted lesion pixels were correctly classified, while the **recall value of 0.881** indicates that most of the actual lesion regions were successfully detected.

To evaluate model robustness, segmentation was also tested under **brightness variation**. The results show a **slight improvement in Dice score (0.891)** and **IoU (0.821)**, suggesting that the model remains stable even when image illumination changes. While **precision decreased slightly to 0.885**, **recall increased to 0.924**, indicating that the model became more sensitive in detecting lesion pixels under varying brightness conditions.

Overall, these results demonstrate that the segmentation model provides **accurate lesion localization with strong overlap between predicted and ground truth masks**, while maintaining robustness under illumination changes. The results also indicate that the model prioritizes **lesion sensitivity**, which is desirable in medical imaging tasks where missing lesion regions can be more critical than slight over-segmentation.

10. Segmentation Error Analysis

Although the segmentation models achieved strong quantitative performance, certain types of prediction errors were observed during qualitative inspection of the results.

One common error type is **over-segmentation**, where the predicted mask slightly extends beyond the actual lesion boundary. This occurs when surrounding skin regions share similar color or texture patterns with the lesion, causing the model to incorrectly include background pixels as part of the lesion.

Another observed issue is **under-segmentation**, where small regions of the lesion are missed by the model. This typically happens when lesion boundaries are faint or when the lesion color closely matches the surrounding skin.

Boundary inaccuracies were also observed in some cases. Dermoscopic images often contain irregular lesion shapes and soft transitions between lesion and skin. This makes it difficult for the model to precisely identify the exact boundary pixels.

Despite these challenges, the overall segmentation performance remained strong. The **Attention U-Net model demonstrated improved sensitivity**, capturing most of the lesion area while maintaining high overlap with the ground truth masks.

Visual overlay analysis showed that the majority of predicted regions correctly overlap with the annotated masks, indicating that the model successfully learned the spatial structure of skin lesions.

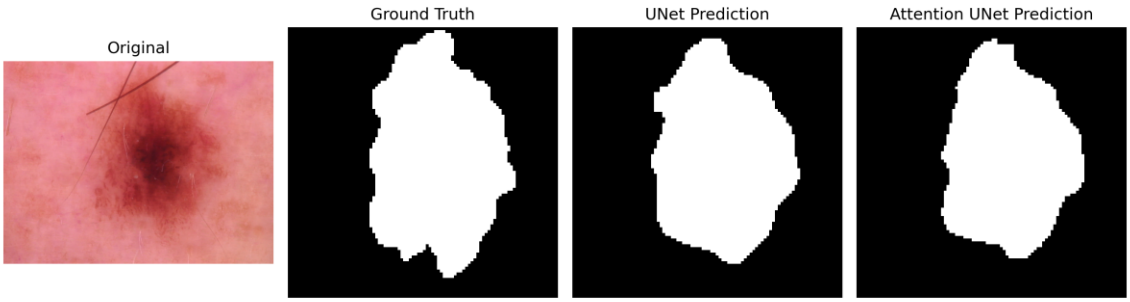

Figure 6: Combined overlay showing Ground Truth (green), Attention U-Net prediction (red), and overlap (yellow).

This figure shows the overlay comparison between the ground truth mask and the prediction generated by the Attention U-Net model. The overlay uses color coding to highlight segmentation

performance: yellow regions indicate correct overlap between predicted and ground truth masks, green regions represent ground truth areas missed by the model, and red regions represent predicted areas outside the true lesion boundary (over-segmentation). The visualization demonstrates that the Attention U-Net model successfully captures the main lesion region with high spatial accuracy, with only minor boundary deviations. This qualitative analysis supports the high Dice score obtained during quantitative evaluation.

9. Conclusion

The experiments conducted in this project demonstrate that deep learning models can effectively perform both **skin disease classification and lesion segmentation** using dermoscopic images.

The classification experiments showed that **ResNet18 outperformed MobileNetV2**, achieving higher overall accuracy due to its deeper architecture and ability to learn richer feature representations. While MobileNetV2 is computationally efficient and suitable for lightweight applications, ResNet18 provided better predictive performance on the HAM10000 dataset.

For segmentation tasks, both **U-Net and Attention U-Net achieved high Dice scores**, indicating strong agreement between predicted masks and ground truth annotations. Attention U-Net demonstrated slightly better lesion sensitivity due to its attention mechanism, which helps the model focus on relevant regions.

The experiments also confirmed that the models maintain stable performance under brightness variation, suggesting that the augmentation strategies used during training improved the model's robustness to illumination changes.

Overall, the combined classification and segmentation pipeline provides a comprehensive framework for automated skin lesion analysis.

Possible Improvements

Several improvements can be considered to further enhance the performance and reliability of the proposed system.

One potential improvement is **training on a more balanced dataset**. The HAM10000 dataset contains significant class imbalance, which can negatively affect classification performance for minority classes. Additional data collection or synthetic data generation could improve model fairness.

Another improvement would be the use of **more advanced architectures**, such as Vision Transformers or hybrid CNN-Transformer models. These architectures have shown promising results in medical imaging tasks due to their ability to capture long-range dependencies in images.

Increasing the **input image resolution** may also improve segmentation accuracy, as higher resolution images preserve more spatial details that help identify lesion boundaries more precisely.

The segmentation model could also benefit from **boundary-aware loss functions** or additional post-processing techniques such as morphological operations to refine predicted lesion edges.

Finally, deploying the system as a **real-time clinical decision-support tool** with an intuitive user interface could significantly enhance its practical usability in dermatology applications.

10. References

The following datasets were used in this project:

HAM10000 Dataset
Human Against Machine with 10000 training images (HAM10000) is a large dermoscopic image dataset widely used for research in automated skin lesion analysis.

Tschandl, P., Rosendahl, C., & Kittler, H. (2018).
The HAM10000 dataset: A large collection of multi-source dermatoscopic images of common pigmented skin lesions.
Scientific Data, 5, 180161.
https://doi.org/10.1038/sdata.2018.161

Dataset Link:
https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000

[2] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in Proc. IEEE CVPR, 2016.

[3] M. Sandler et al., “MobileNetV2: Inverted residuals and linear bottlenecks,” in Proc. IEEE CVPR, 2018.

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